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ImFace: A Nonlinear 3D Morphable Face Model with Implicit Neural Representations

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Abstract

Precise representations of 3D faces are beneficial to various computer vision and graphics applications. Due to the data discretization and model linearity, however, it remains challenging to capture accurate identity and expression clues in current studies. This paper presents a novel 3D morphable face model, namely ImFace, to learn a nonlinear and continuous space with implicit neural representations. It builds two explicitly disentangled deformation fields to model complex shapes associated with identities and expressions, respectively, and designs an improved learning strategy to extend embeddings of expressions to allow more diverse changes. We further introduce a Neural Blend-Field to learn sophisticated details by adaptively blending a series of local fields. In addition to ImFace, an effective preprocessing pipeline is proposed to address the issue of watertight input requirement in implicit representations, enabling them to work with common facial surfaces for the first time. Extensive experiments are performed to demonstrate the superiority of ImFace.

1. Introduction

3D Morphable Face Models (3DMMs) are well-reputed statistical models, established by learning techniques upon prior distributions of facial shapes and textures from a set of samples with dense correspondence, aiming at rendering realistic faces of a high variety. Since a morphable representation is unique across different downstream tasks where the geometry and appearance are separately controllable, 3DMMs are pervasively exploited in many face analysis applications in the field of computer vision, computer graphics, biometrics, and medical imaging [2, 8, 27, 50].

In 3DMMs, the most fundamental issue lies in the way to generate latent morphable representations, and during the past two decades, along with data improvement in scale, diversity and quality [10, 13, 31, 58], remarkable progresses have been achieved. The methods are initially linear model based [7, 39, 40] and further extended to multilinear model based [9, 12, 55], where different modes are individually encoded. Unfortunately, for the relatively limited representation ability of linear models, these methods are not so competent at handling the cases with complicated variations, *e.g.* exaggerated expressions. In the context of deep learning, a number of nonlinear models have been investigated with the input of 2D images [53, 54] or 3D meshes [5, 11, 15, 17, 46] by using Convolutional Neural Networks or Graph Neural Networks. They indeed deliver performance gains; however, restricted by the resolution of discrete representing strategies on input data, facial priors are not sufficiently captured, incurring loss of shape details. Besides, all current methods are dependent on the preposed procedure of point-to-point correspondence [1,6,25,34], but face registration itself remains challenging.

Recently, several studies on Implicit Neural Representations (INRs) [16, 26, 32, 35, 38] have shown that 3D geometries can be precisely modeled by learning continuous deep implicit functions. They describe an input observation as a low-dimensional shape embedding and estimate the Signed

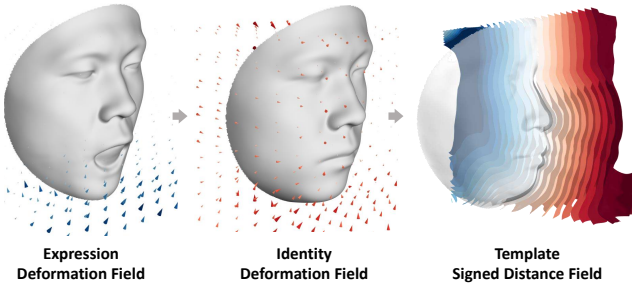


Figure 1. ImFace encodes complex face variations by two explicitly disentangled deformation fields with respect to a template face, resulting in a morphable implicit representation for 3D faces.

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ImFace：基于隐式神经表示的非线性三维可变形人脸模型

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摘要

三维人脸精确表示对多种计算机视觉与图形应用十分有益。然而，由于数据离散化和模型线性化，当前研究在捕捉准确的身份与表情线索方面仍面临挑战。本文提出一种新颖的三维可变形人脸模型，即ImFace，利用隐式神经表示学习非线性且连续的空间。该模型构建两个显式解耦的变形场，分别建模与身份和表情相关的复杂形状，并设计一种改进的学习策略，以扩展表情嵌入，支持更多样化的变化。我们进一步引入神经混合场，通过自适应混合一系列局部场来学习精细细节。除ImFace外，本文还提出一种有效的预处理流程，以解决隐式表示中水密输入要求的问题，首次使其能够处理常见的人脸表面。大量实验验证了ImFace的优越性。

1. 引言

三维可变形人脸模型（3DMMs）是享有盛誉的统计模型，通过从具有密集对应关系的样本集中学习人脸形状与纹理的先验分布而建立，旨在渲染高多样性的逼真人脸。由于可变形表示在不同下游任务中具有独特性，其中几何与外观可分别控制，3DMMs在计算机视觉、计算机图形学、生物识别和医学影像等领域的许多人脸分析应用中被广泛使用[2, 8, 27, 50]。

在3DMM中，最根本的问题在于如何生成潜在的可变形表示，并且在

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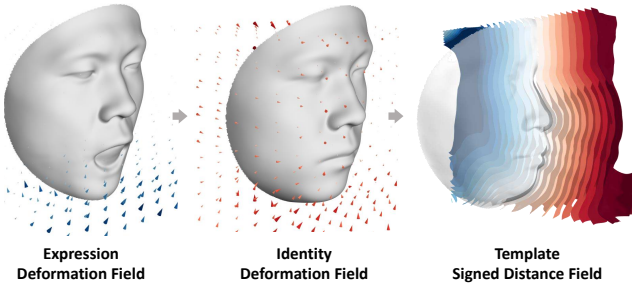


图1：ImFace通过两个相对于模板人脸显式解耦的形变场来编码复杂的人脸变化，从而为3D人脸生成可塑的隐式表示。

过去二十年，随着数据在规模、多样性和质量上的提升[10, 13, 31, 58]，取得了显著进展。这些方法最初基于线性模型[7, 39, 40]，并进一步扩展为基于多线性模型[9, 12, 55]，其中不同模态被分别编码。遗憾的是，由于线性模型的表示能力相对有限，这些方法在处理复杂变化（如夸张表情）时并不十分胜任。在深度学习的背景下，许多非线性模型被研究，输入为2D图像[53, 54]或3D网格[5, 11, 15, 17, 46]，使用卷积神经网络或图神经网络。它们确实带来了性能提升；然而，受限于输入数据离散表示策略的分辨率，面部先验未被充分捕获，导致形状细节丢失。此外，所有现有方法都依赖于预先提出的点对点对应过程[1, 6, 25, 34]，但人脸配准本身仍然具有挑战性。

最近，多项关于隐式神经表示（INRs）[16, 26, 32, 35, 38]的研究表明，通过学习连续深度隐式函数可以精确建模3D几何形状。它们将输入观测描述为低维形状嵌入，并估计符号

Distance Function (SDF) or the occupancy value of a query point so that the surface of an arbitrary resolution and topology can be defined by an isocontour. Due to the continuous parameterization and consistent representation, INRs prove superior to the discrete voxels, pointclouds and meshes, and report decent results in shape reconstruction [24, 56, 57, 61] and surface registration [21, 33, 62]. Such an advantage suggests an alternative to 3DMM that can fulfill accurate correspondence and fine-grained modeling in a unified network. Nevertheless, unlike the objects with apparent shape differences and limited non-rigid variations such as indoor scenes and human bodies, all facial surfaces look very similar but include more complex deformations, where multiple identities and rich expressions deeply interweave with each other, making current INR methods problematic in face modeling, as evidenced by the preliminary attempt [59]. Another difficulty is that implicit functions primarily require watertight input, which is not friendly to facial surfaces.

This paper proposes a novel 3D face morphable model, namely ImFace, which substantially upgrades conventional 3DMMs by learning INRs. To capture nonlinear facial geometry changes, ImFace builds separate INR sub-networks to explicitly disentangle shape morphs into two deformation fields for identity and expression respectively (as Fig. 1 shows), and an improved embedding learning strategy is introduced to extend the latent space of expressions to allow more diverse details. In this way, inter-individual differences and fine-grained deformations can be accurately modeled, which simultaneously takes into account the flexibility when applied to related tasks. Furthermore, inspired by linear blend skinning [30], a Neural Blend-Field is presented to decompose the entire facial deformation or geometry into semantically meaningful regions encoded by a set of local implicit functions and adaptively blend them through a lightweight module, leading to more sophisticated representations with reduced parameters. Besides, a new preprocessing pipeline is designed, which bypasses the need of watertight face data as in existing SDF-based INR models and works well for various facial surfaces, *i.e.*, either hardware-acquired or artificially synthesized.

In summary, the main contributions of this study include:

- We propose a novel INR-based 3DMM, which encodes complex face shape variations by two explicitly disentangled deformation fields, learning powerful representations in a fine-grained and semantically meaningful manner.
- We present an effective preprocessing pipeline, which defines a general SDF for non-watertight 3D faces, enabling INRs to work with them for the first time.
- We experimentally demonstrate that ImFace has the advantage in synthesizing high-quality 3D faces with plausible details, outperforming the state-of-the-art counterparts in 3D face reconstruction.

2. Related Work

3D Morphable Face Models. 3DMMs were first proposed by Blanz and Vetter [7] as a general face representation. A known template mesh was registered to all training scans by utilizing Non-rigid Iterative Closest Point algorithm (NICP) [4], and Principal Components Analysis (PCA) was used to span prior face distributions. To model identity-dependent expressions, 3DMMs were further extended to multilinear models [9, 12, 55]. After then, great advancements have been made with data improvement [10, 13, 31, 58]. FLAME [31] was an expressive model controlling facial expressions by combining jaw articulation with linear expression blendshapes. It was learned from a large 3D face dataset including D3DFACS [20] and delivered more impressive results than ever, whereas nonlinear facial deformations cannot be well captured.

The development of deep networks has catalyzed more powerful nonlinear 3DMMs. A number of models were learned from 2D images [44, 53, 54], but they mostly lacked high fidelity and fine details due to the low resolution of input images in this ill-posed inverse problem. To better leverage 3D face scans, Bagautdinov *et al.* [5] mapped 3D meshes to the 2D space and more studies [11, 15, 17, 46] directly learned 3DMMs from meshes by spectral or spiral convolutions. These neural networks were established upon discrete 3D representations, thus limiting the performance in the presence of complex deformations.

Please refer to [22] for a more comprehensive report on 3DMMs.

Implicit Neural Representations. Recently, implicit neural functions have emerged as a more effective and suitable representation of 3D geometry [16, 26, 32, 35, 38, 49] as they model shapes continuously without discretization. To preserve fine details, the structured local features divided by shape elements [23], grids [28, 42] or octrees [51, 52] were further exploited. Moreover, to well capture shape variations and correspondence relationships, an additional implicit deformation latent space was specially learned [21, 62]. However, high visual fidelity and variety could hardly be simultaneously achieved by existing techniques and they are therefore not so qualified to morphable face modeling. In addition, current studies on INRs mostly focused on the watertight input, such as the ones in ShapeNet [60], and a number of methods on watertight human heads or bodies were proposed accordingly [3, 14, 19, 43, 45, 47, 48, 59]. Among them, H3D-Net [45] learned an implicit head shape space for 2D reconstruction, which was not a generic model. i3DMM [59] was the first implicit 3D morphable model designed for human heads. However, it severely suffered from a low quality in representing facial regions. To overcome the limitation of watertightness, [18] learned an Unsigned Distance Function (UDF) to handle open surfaces, but the reconstructed results were not adequately plausible.

距离函数（SDF）或查询点的占用值，使得任意分辨率和拓扑的表面可以通过等值线来定义。由于连续的参数化和一致的表示，INR在形状重建[24, 56, 57, 61]和表面配准[21,33,62]方面优于离散体素、点云和网格，并取得了不错的结果。这一优势表明，INR可以作为3DMM的替代方案，在统一的网络中实现精确的对应和细粒度建模。然而，与室内场景和人体等具有明显形状差异且非刚性变化有限的对象不同，所有面部表面看起来非常相似，但包含更复杂的变形，其中多个身份和丰富的表情深度交织在一起，使得当前的INR方法在人脸建模中存在问题，初步尝试[59]也证明了这一点。另一个困难是隐式函数主要要求水密输入，这对面部表面并不友好。

本文提出了一种新颖的三维人脸可变形模型，命名为ImFace，通过学习隐式神经表示（INRs）大幅提升了传统三维可变形模型（3DMMs）的性能。为捕捉非线性的面部几何变化，ImFace构建了独立的INR子网络，将形状变形显式解耦为分别对应身份和表情的两个变形场（如图1所示），并引入了一种改进的嵌入学习策略，以扩展表情的潜在空间，从而允许更丰富的细节表达。通过这种方式，个体间差异和细粒度变形可以被精确建模，同时兼顾了在相关任务中应用的灵活性。此外，受线性混合蒙皮[30]的启发，本文提出了一种神经混合场（Neural Blend-Field），将整个人脸变形或几何分解为若干由局部隐函数编码的语义上有意义的区域，并通过一个轻量级模块自适应地混合这些区域，从而在减少参数的同时实现更复杂的表示。另外，本文还设计了一种新的预处理流程，该流程无需像现有基于SDF的INR模型那样要求水密人脸数据，并且适用于各种人脸表面，无论是硬件采集的还是人工合成的。

总之，本研究的主要贡献包括：

- 我们提出了一种基于INR的新型3DMM，通过两个显式解耦的形变场编码复杂的人脸形状变化，以细粒度且语义上有意义的方式学习强大的表示。
- 我们提出了一种有效的预处理流程，该流程为非水密三维人脸定义了通用SDF，首次使INR能够处理此类数据。
- 我们通过实验证明，ImFace在合成具有合理细节的高质量3D人脸方面具有优势，在3D人脸重建中优于最先进的同类方法。

2. 相关工作

3D可变形人脸模型。3DMM最早由Blanz和Vetter [7]提出，作为一种通用的人脸表示方法。通过利用非刚性迭代最近点算法（NICP）[4]将已知模板网格配准到所有训练扫描数据上，并采用主成分分析（PCA）来构建先验人脸分布。为了建模与身份相关的表情，3DMM进一步扩展为多线性模型[9, 12, 55]。此后，随着数据质量的提升[10, 13, 31, 58]，该领域取得了重大进展。FLAME [31]是一种富有表现力的模型，通过将下颌关节运动与线性表情混合形状相结合来控制面部表情。该模型基于包含D3DFACS [20]的大规模3D人脸数据集训练而成，取得了前所未有的显著效果，但仍无法很好地捕捉非线性面部形变。

深度网络的发展催生了更强大的非线性3DMM。许多模型从2D图像中学习得到[44, 53, 54]，但由于这一病态逆问题中输入图像分辨率较低，它们大多缺乏高保真度和精细细节。为了更好地利用3D人脸扫描数据，Bagautdinov等人[5]将3D网格映射到2D空间，更多研究[11, 15, 17, 46]则通过谱卷积或螺旋卷积直接从网格中学习3DMM。这些神经网络建立在离散3D表示之上，因此在面对复杂变形时性能受限。

关于3DMMs的更全面报告，请参阅[22]。

隐式神经表示。近年来，隐式神经函数已成为三维几何更有效且更合适的表示方式[16,26,32,35,38,49]，因为它们无需离散化即可连续地对形状进行建模。为了保留精细细节，进一步利用了按形状元素[23]、网格[28,42]或八叉树[51,52]划分的结构化局部特征。此外，为了很好地捕捉形状变化和对应关系，专门学习了额外的隐式变形潜空间[21,62]。然而，现有技术难以同时实现高视觉保真度和多样性，因此它们不太适合可变形人脸建模。此外，当前关于隐式神经表示的研究大多集中于水密输入，例如ShapeNet[60]中的输入，并相应提出了许多针对水密人头或人体的方法[3,14,19,43,45,47,48,59]。其中，H3D-Net[45]学习了用于二维重建的隐式头部形状空间，但该模型并非通用模型。i3DMM[59]是首个专为人头设计的隐式三维可变形模型。然而，它在表示面部区域方面存在严重的低质量问题。为了克服水密性的限制，[18]学习了无符号距离函数（UDF）来处理开放表面，但重建结果不够合理。

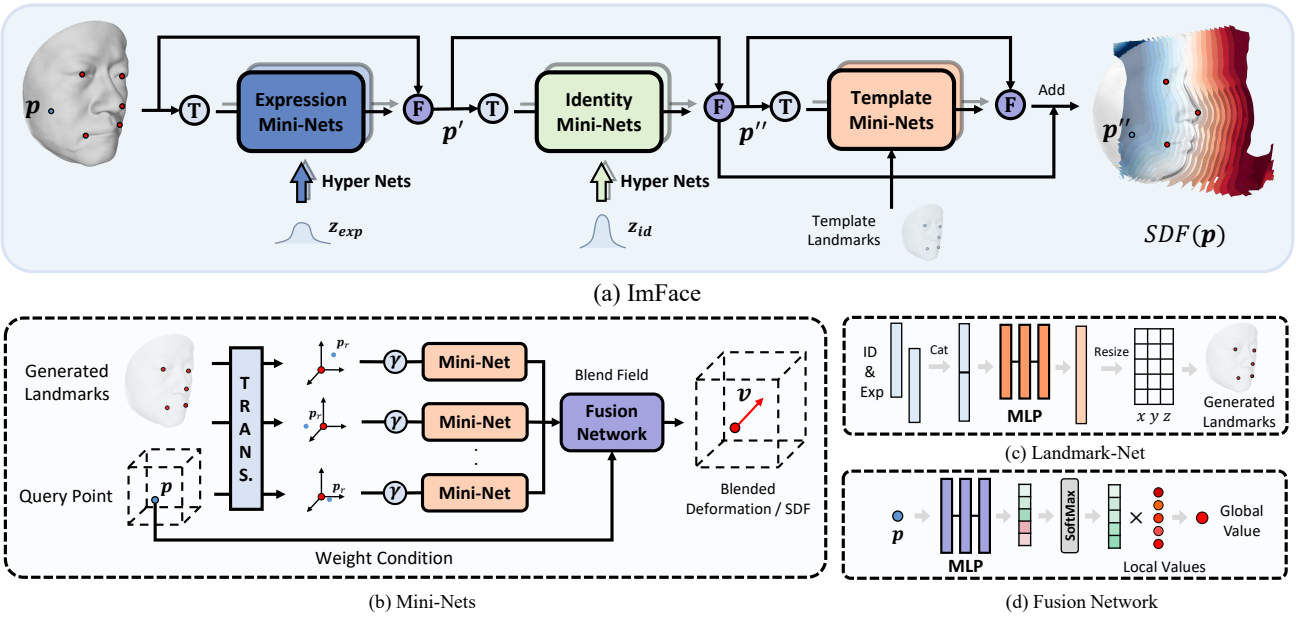


Figure 2. **ImFace overview.** **(a)** The proposed network consists of three Mini-Nets blocks to explicitly disentangle shape morphs into separate deformation fields, where the Expression and Identity Mini-Nets blocks are associated with expression and identity deformations, respectively, and the Template Mini-Nets block learns the SDF of a template face space. **(b)** The Mini-Nets block is a shared architecture, which decomposes an entire facial feature into semantically meaningful parts and encodes them by a set of local field functions. It is tailed by a Fusion Network for more comprehensive representations. **(c)** The Landmark-Net is introduced to softly partition the entire facial surface. **(d)** The Fusion Network is a lightweight module conditioned on the query point position, which adaptively blends the local field functions, resulting in an elaborate Neural Blend-Field.

3. Method

We take advantages of INRs to learn a nonlinear 3D morphable face model. The proposed ImFace explicitly disentangles facial shape morphs into two separate deformation fields associated with identity and expression, respectively, and a deep SDF is learned to represent the template shape. All the fields are blended with a series of local implicit functions for more detailed representations.

3.1. Disentangled INRs Network

The fundamental idea of INRs is to train a neural network to fit a continuous function f, which implicitly represents surfaces through level-sets. The function can be defined in various formats, *e.g.* occupancies [35], SDF [38], or UDF [18]. We exploit a deep SDF conditioned on the latent embeddings of both expression and identity for comprehensive face representations. It outputs the signed distance s from a query point:

$$f : (\mathbf{p}, \mathbf{z}_{exp}, \mathbf{z}_{id}) \in \mathbb{R}^3 \times \mathbb{R}^{d_{exp}} \times \mathbb{R}^{d_{id}} \mapsto s \in \mathbb{R}, \quad (1)$$

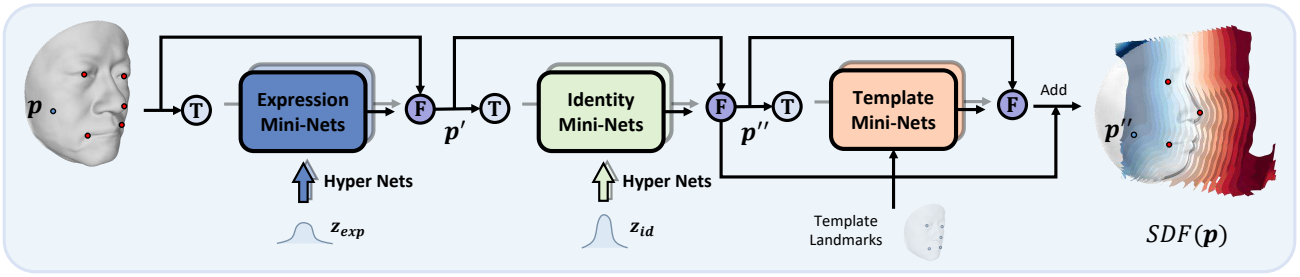
where $\mathbf{p} \in \mathbb{R}^3$ is the coordinate of the query point in the 3D space, $\mathbf{z}_{exp}$ and $\mathbf{z}_{id}$ denote the expression and identity embeddings, respectively.

Our goal is to learn a neural network to parameterize f, making it satisfy the genuine facial shape priors. As shown in Fig. 2, the proposed network for Imface is composed of three Mini-Nets blocks, which explicitly disentangles the learning process of facial shape morphs, ensuring that inter-individual differences and fine-grained deformations can be accurately modeled. In particular, the first two Mini-Nets blocks learn separate deformation fields associated with expression- and identity-variation, respectively, and the Template Mini-Nets block learns a signed distance field of a template face shape.

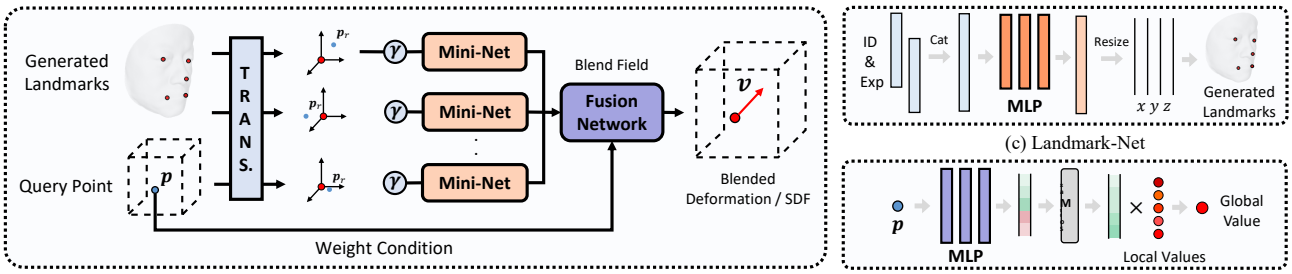
All the fields above are implemented by a shared Mini-Nets architecture, where the entire facial deformation or geometry is further decomposed into a number of semantically meaningful parts and encoded by a set of local field functions, so that rich details can be sufficiently captured. A lightweight module conditioned on the query point position, *i.e.,* Fusion Network, is stacked at the end of the Mini-Nets block to adaptively blend the local fields. As such, an elaborate Neural Blend-Field is achieved. The three core components of ImFace work for different purposes and their structures are slightly changed accordingly. We briefly describe them as follows:

Expression Mini-Nets (ExpNet). The facial deformations incurred by expressions are represented by ExpNet $\mathcal{E}$, which learns an observation-to-canonical warping for every face scan:

$$\mathcal{E} : (\mathbf{p}, \mathbf{z}_{exp}, l) \mapsto \mathbf{p}' \in \mathbb{R}^3, \quad (2)$$



(a) ImFace



(b) Mini-Nets

(c) Landmark-Net

(d) Fusion Network

图2. ImFace概述。(a) 所提出的网络由三个Mini-Nets块组成，用于将形状变形显式解耦为独立的变形场，其中表情和身份Mini-Nets块分别与表情变形和身份变形相关联，而模板Mini-Nets块学习模板面部空间的SDF。(b) Mini-Nets块是一种共享架构，它将整个面部特征分解为语义上有意义的部分，并通过一组局部场函数对其进行编码。其尾部连接一个融合网络，以获得更全面的表示。(c) 引入Landmark-Net以柔和地划分整个面部表面。(d) 融合网络是一个以查询点位置为条件的轻量级模块，它自适应地混合局部场函数，从而产生精细的神经混合场。

3. 方法

我们利用INR的优势来学习一个非线性的3D可变形人脸模型。所提出的ImFace将面部形状变形显式解耦为分别与身份和表情相关的两个独立变形场，并学习一个深度SDF来表示模板形状。所有场通过一系列局部隐式函数进行融合，以获得更精细的表示。

3.1. 解耦INRs网络

INR的基本思想是训练一个神经网络来拟合连续函数f，该函数通过水平集隐式表示表面。该函数可以以多种格式定义，例如占用率[35]、SDF[38]或UDF[18]。我们利用一个以表情和身份的潜在嵌入为条件的深度SDF来实现全面的面部表示。它输出查询点的有符号距离s：

$$f:(\mathbf{p},\mathbf{z}_{exp},\mathbf{z}_{id})\in\mathbb{R}^3\times\mathbb{R}^{d_{exp}}\times\mathbb{R}^{d_{id}}\mapsto s\in\mathbb{R}, \quad (1)$$

其中 $\mathbf{p}\in\mathbb{R}^3$ 是查询点在三维空间中的坐标，$\mathbf{z}_{exp}$ 和 $\mathbf{z}_{id}$ 分别表示表情嵌入和身份嵌入。

我们的目标是学习一个神经网络来参数化f，使其满足真实的人脸形状先验。由于如图2所示，所提出的Imface网络由三个Mini-Nets块组成，明确解耦了面部形状变形的学习过程，确保个体间差异和细粒度形变能够被精确建模。具体而言，前两个Mini-Nets块分别学习与表情变化和身份变化相关的独立变形场，而模板Mini-Nets块学习模板面部形状的有符号距离场。

上述所有场均由一个共享的Mini-Nets架构实现，其中整个面部形变或几何结构被进一步分解为若干语义上有意义的部件，并通过一组局部场函数进行编码，从而充分捕获丰富细节。一个以查询点位置为条件的轻量级模块，即融合网络，堆叠在Mini-Nets块的末端，用于自适应地混合局部场。由此，一个精细的神经混合场得以实现。ImFace的三个核心组件服务于不同目的，其结构也相应略有变化。我们简要描述如下：

表达式微型网络（ExpNet）。由表情引起的面部形变由ExpNet $\mathcal{E}$表示，该网络学习每个面部扫描的观测到规范空间的扭曲映射：

$$\mathcal{E}:(\mathbf{p},\mathbf{z}_{exp},l)\mapsto\mathbf{p}'\in\mathbb{R}^3, \quad (2)$$

where $l \in \mathbb{R}^{k\times 3}$ denotes k 3D landmarks on an observed face generated by a Landmark-Net $\eta : (\mathbf{z}_{exp}, \mathbf{z}_{id}) \mapsto l$, introduced to localize the query point $\mathbf{p}$ in the Neural Blend-Field. The point $\mathbf{p}$ in the observation space is deformed by $\mathcal{E}$ to a new point $\mathbf{p}'$ in the person-specific canonical space, which represents faces with a neutral expression.

Identity Mini-Nets (IDNet). To model shape morphs among individuals, IDNet $\mathcal{I}$ further warps the canonical space to a template shape space shared by all faces:

$$\mathcal{I} : (\mathbf{p}', \mathbf{z}_{id}, l') \mapsto (\mathbf{p}'', \delta) \in \mathbb{R}^3 \times \mathbb{R}, \tag{3}$$

where $l' \in \mathbb{R}^{k\times 3}$ denotes k landmarks on the canonical face generated by another Landmark-Net conditioned only on the identity embedding $\eta' : \mathbf{z}_{id} \mapsto l'$, and $\mathbf{p}''$ is the deformed point in the template space. To cope with the possible non-existent correspondences generated during preprocessing, $\mathcal{I}$ additionally predicts a residual term $\delta \in \mathbb{R}$ to correct the predicted SDF value s_0, which is similar to [21].

Template Mini-Nets (TempNet). TempNet $\mathcal{T}$ learns a signed distance field of the shared template face:

$$\mathcal{T} : (\mathbf{p}'', l'') \mapsto s_0 \in \mathbb{R}, \tag{4}$$

where $l'' \in \mathbb{R}^{k\times 3}$ denotes k landmarks on the template face, which is averaged on the whole training set, and s_0 is the uncorrected SDF value. The final SDF value of a query point is calculated via $s = s_0 + \delta$, and the ImFace model can be ultimately formulated as:

$$f(\mathbf{p}) = \mathcal{T}(\mathcal{I}_{\mathbf{p}''}(\mathcal{E}(\mathbf{p}, \mathbf{z}_{exp}), \mathbf{z}_{id})) + \mathcal{I}_\delta(\mathcal{E}(\mathbf{p}, \mathbf{z}_{exp}), \mathbf{z}_{id}). \tag{5}$$

The proposed ImFace learns face morphs by disentangled deformation fields in a fine-grained and meaningful manner, ensuring that more diverse and sophisticated facial deformations can be accurately learned. We detailedly introduce the architectures of the main modules, learning strategy, training critics and data preprocessing pipeline in the subsequent.

3.2. Neural Blend-Field

The Mini-Nets block is a common architecture shared by the three sub-networks $\mathcal{E}$, $\mathcal{I}$, and $\mathcal{T}$. It learns a continuous field function $\boldsymbol{\psi} : \mathbf{x} \in \mathbb{R}^3 \mapsto v$, to produce a Neural Blend-Field for comprehensive face representations. In particular, to overcome the limited expressivity of a single network, we decompose a face space into a set of semantically meaningful local regions, and learn v (*e.g.* deformation or signed distance value) individually before blending. Such design is inspired by the recent INRs study [41] on human body, which introduces the linear blend skinning algorithm [30] to make the network learn from separate transformations of body parts. To better represent detailed facial surfaces, we replace the constant transformation term in the original linear blend skinning algorithm with $\boldsymbol{\psi}_n(\mathbf{x} - l_n)$, and define the Neural Blend-Field as:

$$v = \boldsymbol{\psi}(\mathbf{x}) = \sum_{n=1}^{k} w_n(\mathbf{x})\boldsymbol{\psi}_n(\mathbf{x} - l_n), \tag{6}$$

where l_n is a parameter that describes the n-th local region, $w_n(\mathbf{x})$ is the n-th blend weight, and $\boldsymbol{\psi}_n(\mathbf{x}-l_n)$ is the corresponding local field. In this way, the blending is performed on a series of local fields, rather than calculating a weighted average of the output values v of some fixed positions, leading to more powerful representation capability in handling complicated local features.

Specifically, five landmarks located at the outer eye corners, mouth corners, and nose tip are utilized to describe the local regions $(l_n \in \mathbb{R}^3)_{n=1}^5$, and each region is assigned a tiny MLP with sinusoidal activations [49] to generate the local field, denoted as $\boldsymbol{\psi}_n$. To capture high-frequency local variations, we leverage sinusoidal positional encoding γ [36] on the coordinate $\mathbf{x} - l_n$. At the end of a Mini-Nets block, a lightweight Fusion Network conditioned on the absolute coordinate of input $\mathbf{x}$ is equipped, which is implemented by a 3-layer MLP with softmax to predict the blend weights $(w_n \in \mathbb{R}^+)_{n=1}^5$.

Deformation Formulation. We formulate the deformation with a SE(3) field $(\boldsymbol{\omega}, \mathbf{v}) \in \mathbb{R}^6$, where $\boldsymbol{\omega} \in so(3)$ is a rotate vector representing the screw axis and the angle of rotation. The deformed coordinates $\mathbf{x}'$ can be calculated by $e^{\boldsymbol{\omega}}\mathbf{x} + \mathbf{t}$, where the rotation matrix $e^{\boldsymbol{\omega}}$ (exponential map form of Rodrigues' formula) is written as:

$$e^{\boldsymbol{\omega}} = \mathbf{I} + \frac{\sin\|\boldsymbol{\omega}\|}{\|\boldsymbol{\omega}\|}\boldsymbol{\omega}^\wedge + \frac{1-\cos\|\boldsymbol{\omega}\|}{\|\boldsymbol{\omega}\|^2}(\boldsymbol{\omega}^\wedge)^2, \tag{7}$$

and the translation $\mathbf{t}$ is formulated as:

$$\mathbf{t} = \left[\mathbf{I} + \frac{1-\cos\|\boldsymbol{\omega}\|}{\|\boldsymbol{\omega}\|^2}\boldsymbol{\omega}^\wedge + \frac{\|\boldsymbol{\omega}\| - \sin\|\boldsymbol{\omega}\|}{\|\boldsymbol{\omega}\|^3}(\boldsymbol{\omega}^\wedge)^2\right]\mathbf{v}, \tag{8}$$

where $\boldsymbol{\omega}^\wedge$ denotes the skew-symmetric matrix of $\boldsymbol{\omega}$. We exploit SE(3) to describe facial shape morphs for its superior capability in handling mandibular rotations and higher robustness to pose perturbations than the common translation deformation $\mathbf{x}' = \mathbf{x} + \mathbf{t}$.

Hyper Nets. To obtain a more compact and expressive latent space, we then introduce a meta-learning approach [49]. A Hyper Net ϕ_n is implemented by an MLP and it predicts the instance-specific parameters for ExpNet $\mathcal{E}$ and IDNet $\mathcal{I}$. It takes a latent code $\mathbf{z}$ as input and generates the parameters for the neurons in a Mini-Net $\boldsymbol{\psi}_n$ so that the learned facial representations possess a higher variety.

3.3. Improved Expression Embedding Learning

The auto-decoder framework proposed by [38] has been widely adopted in INRs to jointly learn embeddings and network parameters. In the previous attempt [59], each expression type is encoded by one embedding for attribute disentangling. Unfortunately, such an embedding is merely able to represent the average shape morph of an expression

其中 $l \in \mathbb{R}^{k\times 3}$ 表示由 Landmark-Net $\eta : (\mathbf{z}_{exp}, \mathbf{z}_{id}) \mapsto l$ 生成的观测人脸 k 个 3D 地标，该网络用于在神经混合场中定位查询点 $\mathbf{p}$。观测空间中的点 $\mathbf{p}$ 经 $\mathcal{E}$ 变形为个体特定规范空间中的新点 $\mathbf{p}'$，该空间以中性表情表示人脸。

身份微型网络（IDNet）。为了对个体间的形状变形进行建模，IDNet $\mathcal{I}$ 进一步将规范空间扭曲到所有面部共享的模板形状空间：

$$\mathcal{I} : (\mathbf{p}', \mathbf{z}_{id}, l') \mapsto (\mathbf{p}'', \delta) \in \mathbb{R}^3 \times \mathbb{R}, \tag{3}$$

其中 k 表示由另一个仅以身份嵌入 η' 为条件的 Landmark-Net 在规范人脸上生成的 k 个地标：$\mathbf{z}_{id} \mapsto l'$，而 $\mathbf{p}''$ 是模板空间中的形变点。为应对预处理过程中可能产生的不存在对应关系，$\mathcal{I}$ 额外预测一个残差项 $\delta \in \mathbb{R}$ 来修正预测的 SDF 值 s_0，这与 [21] 类似。模板迷你网络（TempNet）。TempNet $\mathcal{T}$ 学习共享模板人脸的符号距离场：

$$\mathcal{T} : (\mathbf{p}'', l'') \mapsto s_0 \in \mathbb{R}, \tag{4}$$

其中 $l'' \in \mathbb{R}^{k\times 3}$ 表示模板面部上的 k 个地标，该模板面部在整个训练集上取平均，而 s_0 是未校正的SDF值。查询点的最终SDF值通过 $s = s_0 + \delta$ 计算，ImFace模型最终可表述为：

$$f(\mathbf{p}) = \mathcal{T}(\mathcal{I}_{\mathbf{p}''}(\mathcal{E}(\mathbf{p}, \mathbf{z}_{exp}), \mathbf{z}_{id})) + \mathcal{I}_{\delta}(\mathcal{E}(\mathbf{p}, \mathbf{z}_{exp}), \mathbf{z}_{id}). \tag{5}$$

所提出的ImFace通过细粒度且有意义的解耦变形场学习面部形变，确保能够准确学习更多样化、更精细的面部变形。下文将详细介绍主要模块的架构、学习策略、训练评判器以及数据预处理流程。

3.2. 神经混合场

Mini-Nets块是三个子网络$\mathcal{E}$、$\mathcal{I}$和$\mathcal{T}$共享的通用架构。它学习一个连续场函数$\boldsymbol{\psi}$：$\mathbf{x} \in \mathbb{R}^3 \mapsto v$，以生成神经混合场，用于全面的面部表示。特别地，为了克服单一网络表达能力的局限性，我们将面部空间分解为一组具有语义意义的局部区域，并分别学习v（（例如形变或符号距离值）），然后再进行混合。该设计受到近期关于人体INRs研究[41]的启发，该研究引入了线性混合蒙皮算法[30]，使网络能够从身体各部分的独立变换中学习。为了更好地表示精细的面部表面，我们将原始线性混合蒙皮算法中的常数变换项替换为 $\boldsymbol{\psi}_n(\mathbf{x} - l_n)$，并将神经混合场定义为：

$$v = \boldsymbol{\psi}(\mathbf{x}) = \sum_{n=1}^{k} w_n(\mathbf{x})\boldsymbol{\psi}_n(\mathbf{x} - l_n), \tag{6}$$

其中 l_n 是描述第 n 个局部区域的参数，$w_n(\mathbf{x})$ 是第 n 个混合权重，$\boldsymbol{\psi}_n(\mathbf{x}-l_n)$ 是对应的局部场。这样，混合操作在一系列局部场上进行，而非计算某些固定位置输出值 v 的加权平均，从而在处理复杂局部特征时具有更强的表示能力。

具体而言，利用位于外眼角、嘴角和鼻尖的五个关键点来描述局部区域（$l_n \in \mathbb{R}^3$）$_{n=1}^{5}$，每个区域分配一个带有正弦激活函数[49]的微型MLP以生成局部场，记为$\boldsymbol{\psi}_n$。为捕捉高频局部变化，我们在坐标$\mathbf{x} - l_n$上采用正弦位置编码γ[36]。在Mini-Nets块末端，配备了一个以输入绝对坐标$\mathbf{x}$为条件的轻量级融合网络，该网络通过一个带softmax的3层MLP实现，用于预测混合权重（$w_n \in \mathbb{R}^+$）^{5}。

$_{n=1}^{\cdot}$变形公式。我们使用SE(3)场$(\boldsymbol{\omega}, \mathbf{v}) \in \mathbb{R}^6$来表述变形，其中$\boldsymbol{\omega} \in so(3)$是表示螺旋轴和旋转角度的旋转向量。变形后的坐标$\mathbf{x}'$可通过$e^{\boldsymbol{\omega}}\mathbf{x} + \mathbf{t}$计算，其中旋转矩阵 $e^{\boldsymbol{\omega}}$ (罗德里格斯公式的指数映射形式)写作：

$$e^{\boldsymbol{\omega}} = \mathbf{I} + \frac{\sin\|\boldsymbol{\omega}\|}{\|\boldsymbol{\omega}\|}\boldsymbol{\omega}^{\wedge} + \frac{1-\cos\|\boldsymbol{\omega}\|}{\|\boldsymbol{\omega}\|^2}(\boldsymbol{\omega}^{\wedge})^2, \tag{7}$$

且翻译 $\mathbf{t}$ 被表述为：

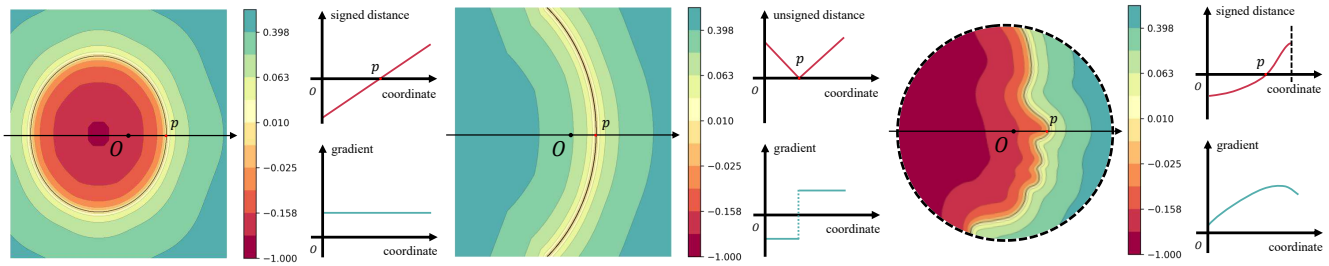
$$\mathbf{t} = \left[\mathbf{I} + \frac{1-\cos\|\boldsymbol{\omega}\|}{\|\boldsymbol{\omega}\|^2}\boldsymbol{\omega}^{\wedge} + \frac{\|\boldsymbol{\omega}\| - \sin\|\boldsymbol{\omega}\|}{\|\boldsymbol{\omega}\|^3}(\boldsymbol{\omega}^{\wedge})^2\right]\mathbf{v}, \tag{8}$$

其中 $\boldsymbol{\omega}^{\wedge}$ 表示 $\boldsymbol{\omega}$ 的反对称矩阵。我们利用 SE(3) 来描述面部形状变形，因为其在处理下颌旋转方面具有优越能力，并且相比常见的平移变形 $\mathbf{x}' = \mathbf{x} + \mathbf{t}$，对姿态扰动具有更高的鲁棒性。

超网络。为了获得更紧凑且更具表达力的潜在空间，我们进一步引入了一种元学习方法[49]。超网络ϕ_n由多层感知机实现，用于预测ExpNet$\mathcal{E}$和IDNet$\mathcal{I}$的实例特定参数。它以潜在编码$\mathbf{z}$为输入，为Mini-Net$\boldsymbol{\psi}_n$中的神经元生成参数，从而使学习到的人脸表示具有更高的多样性。

3.3. 改进的表达式嵌入学习

[38]提出的自动解码器框架已被广泛采用于隐式神经表示（INRs）中，用于联合学习嵌入和网络参数。在先前的工作[59]中，每种表情类型通过一个嵌入进行编码以实现属性解耦。遗憾的是，这样的嵌入仅能表示表情的平均形状变形。



(a) Signed Distance Field (b) Unsigned Distance Field (c) SDF on pseudo watertight face

Figure 3. **(a)** SDF is able to represent closed shapes. **(b)** UDF is capable of representing an open surface but the gradient is discontinuous at the boundary, making it hard to be fitted by neural networks. **(c)** The proposed method generates pseudo watertight faces and restricts implicit functions on them, enabling implicit neural networks to learn geometry representations on 3D faces.

type, making the learned latent space fail to capture more diverse deformation details across individuals. To avoid the dilemma above, we improve the learning strategy by treating each non-neutral face scan as a unique expression and generating a specific embedding for it. In this way, the latent space is significantly extended, which enables $\mathcal{E}$ to represent more fine-grained details. On the other side, there exists a potential failure mode that the identity properties are tangled into the expression space again, and $\mathcal{I}$ collapses to an identity mapping. To tackle this challenge, we suppress $\mathcal{E}$ when the current training sample is a neutral face, written as:

$$\mathcal{E}(\mathbf{p}_{nu}, \mathbf{z}_{exp}, l) \equiv \mathbf{p}_{nu}, \tag{9}$$

where $\mathbf{p}_{nu}$ denotes a point from a neutral face. By applying such a learning strategy, $\mathcal{I}$ and $\mathcal{T}$ jointly learn shape representations on neutral faces, and $\mathcal{E}$ focuses only on expression deformations. Moreover, only neutral labels are required during training, bypassing the dense expression labels.

3.4. Loss Functions

ImFace is trained with several loss functions to learn plausible facial shape representations and dense correspondence.

Reconstruction Loss. The basic SDF structure loss is applied to learn implicit fields:

$$\mathcal{L}^i_{sdf} = \lambda_1 \sum_{\mathbf{p}\in\Omega_i} |f(\mathbf{p}) - \bar{s}| + \lambda_2 \sum_{\mathbf{p}\in\Omega_i} (1 - \langle \nabla f(\mathbf{p}), \bar{\mathbf{n}} \rangle), \tag{10}$$

where $\bar{s}$ and $\bar{\mathbf{n}}$ denote the ground-truth SDF values and the field gradients, respectively. Ω_i is the sampling space of the face scan i, and λ indicates the trade-off parameter.

Eikonal Loss. To obtain reasonable fields throughout the network, multiple Eikonal losses are used to enforce the L-2 norm of spatial gradients to be unit:

$$\mathcal{L}^i_{eik} = \lambda_3 \sum_{\mathbf{p}\in\Omega_i} \left(|\|\nabla f(\mathbf{p})\| - 1| + |\|\nabla \mathcal{T}(\mathcal{I}(\mathbf{p}'))\| - 1| \right), \tag{11}$$

where $\mathcal{L}^i_{eik}$ enables the network to satisfy the Eikonal constraint [26] in the observation and canonical spaces simultaneously, which also contributes to a reasonable correspondence along face deformations at all the network stages.

Embedding Loss. It regularizes the embedding with a zero-mean Gaussian prior:

$$\mathcal{L}^i_{emb} = \lambda_4 \left(\|\mathbf{z}_{exp}\|^2 + \|\mathbf{z}_{id}\|^2 \right). \tag{12}$$

Landmark Generation Loss. The l_1-loss is used to learn the Landmark-Nets η, η':

$$\mathcal{L}^i_{lmk_g} = \lambda_5 \sum_{n=1}^{k} \left(|l_n - \bar{l^i_n}| + |l'_n - \bar{l'_n}| \right), \tag{13}$$

where $\bar{l^i}$ denotes the k labeled landmarks on sample i, and $\bar{l'}$ denotes the landmarks on the corresponding neutral face.

Landmark Consistency Loss. We exploit this loss to guide the deformed landmarks to be located at the corresponding positions on the ground-truth neutral and template faces for better correspondence performance:

$$\mathcal{L}^i_{lmk_c} = \lambda_6 \sum_{n=1}^{k} \left(|\mathcal{E}(l_n) - \bar{l'}_n| + |\mathcal{I}(\mathcal{E}(l_n)) - l''_n| \right). \tag{14}$$

Residual Constraint. As in [21], to avoid that the residual item δ learns too much template face information and downgrades the morphable model, we penalize δ by:

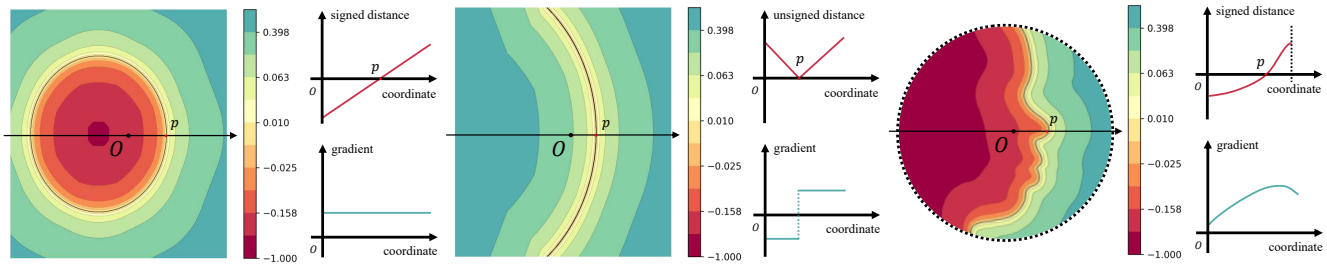
$$\mathcal{L}^i_{res} = \lambda_7 \sum_{\mathbf{p}\in\Omega_i} |\delta(\mathbf{p})|. \tag{15}$$

The total training loss is calculated on all face samples indexed by i, finally formulated as:

$$\mathcal{L} = \sum_i (\mathcal{L}^i_{sdf} + \mathcal{L}^i_{eik} + \mathcal{L}^i_{emb} + \mathcal{L}^i_{lmk_g} + \mathcal{L}^i_{lmk_c} + \mathcal{L}^i_{res}). \tag{16}$$

At the testing phase, for each 3D face indexed by j, we minimize the following objective to obtain its latent embedding and the reconstructed 3D face:

$$\arg\min_{\mathbf{z}_{exp}, \mathbf{z}_{id}} \sum_j (\mathcal{L}^j_{sdf} + \mathcal{L}^j_{eik} + \mathcal{L}^j_{emb}). \tag{17}$$



(a) Signed Distance Field (b) Unsigned Distance Field (c) SDF on pseudo watertight face

图3. (a) SDF能够表示封闭形状。(b) UDF能够表示开放表面，但在边界处梯度不连续，使其难以被神经网络拟合。(c) 所提出的方法生成伪水密面并限制隐式函数在其上，使隐式神经网络能够在3D面上学习几何表示。

类型，使得学习到的潜在空间无法捕捉跨个体的更多样化形变细节。为避免上述困境，我们改进了学习策略，将每个非中性面部扫描视为一种独特表情，并为其生成特定的嵌入。这样一来，潜在空间得到显著扩展，使$\mathcal{E}$能够表示更细粒度的细节。另一方面，存在一种潜在的失败模式，即身份属性再次纠缠到表情空间中，且$\mathcal{I}$退化为恒等映射。为应对这一挑战，当当前训练样本为中性面部时，我们抑制$\mathcal{E}$，表示为：

$$\mathcal{E}(\mathbf{p}_{nu}, \mathbf{z}_{exp}, l) \equiv \mathbf{p}_{nu}, \tag{9}$$

其中$\mathbf{p}_{nu}$表示来自中性面部的一个点。通过应用这种学习策略，$\mathcal{I}$和$\mathcal{T}$共同学习中性面部上的形状表示，而$\mathcal{E}$仅关注表情形变。此外，训练过程中仅需要中性标签，无需密集的表情标签。

3.4. 损失函数

ImFace通过多种损失函数进行训练，以学习合理的面部形状表示和密集对应关系。

重建损失。应用基本的SDF结构损失来学习隐式场：

$$\mathcal{L}^i_{sdf} = \lambda_1 \sum_{\mathbf{p}\in\Omega_i} |f(\mathbf{p}) - \bar{s}| + \lambda_2 \sum_{\mathbf{p}\in\Omega_i} (1 - \langle \nabla f(\mathbf{p}), \bar{\mathbf{n}} \rangle), \tag{10}$$

其中 $\bar{s}$ 和 $\bar{\mathbf{n}}$ 分别表示真实SDF值和场梯度。Ω_i 是人脸扫描 i 的采样空间，λ 表示权衡参数。

Eikonal损失。为了在整个网络中获得合理的场，使用多个Eikonal损失来强制空间梯度的L-2范数为单位：

$$\mathcal{L}^i_{eik} = \lambda_3 \sum_{\mathbf{p}\in\Omega_i} \left(|\|\nabla f(\mathbf{p})\| - 1| + |\|\nabla \mathcal{T}(\mathcal{I}(\mathbf{p}'))\| - 1| \right), \tag{11}$$

其中$\mathcal{L}^i_{eik}$使网络能够在观测空间和规范空间中同时满足Eikonal约束[26]，同时，这也促进了网络各阶段面部形变过程中合理的对应关系。

嵌入损失。它使用零均值高斯先验对嵌入进行正则化：

$$\mathcal{L}^i_{emb} = \lambda_4 \left(\|\mathbf{z}_{exp}\|^2 + \|\mathbf{z}_{id}\|^2 \right). \tag{12}$$

地标生成损失。使用l_1-损失来学习地标网络η、η'：

$$\mathcal{L}^i_{lmk_g} = \lambda_5 \sum_{n=1}^{k} \left(|l_n - \bar{l^i_n}| + |l'_n - \bar{l'_n}| \right), \tag{13}$$

其中 $\bar{l^i}$ 表示样本 i 上标记为 k 的标注地标点，$\bar{l'}$ 表示对应中性面部上的地标点。地标一致性损失。我们利用该损失引导变形后的地标点位于真实中性面部和模板面部上的对应位置，以获得更好的对应性能：

$$\mathcal{L}^i_{lmk_c} = \lambda_6 \sum_{n=1}^{k} \left(|\mathcal{E}(l_n) - \bar{l'}_n| + |\mathcal{I}(\mathcal{E}(l_n)) - l''_n| \right). \tag{14}$$

残差约束。如文献[21]所述，为避免残差项δ学习过多模板人脸信息而降低可变形模型的性能，我们通过以下方式对δ进行惩罚：

$$\mathcal{L}^i_{res} = \lambda_7 \sum_{\mathbf{p}\in\Omega_i} |\delta(\mathbf{p})|. \tag{15}$$

总训练损失在由i索引的所有面部样本上计算，最终公式化为：

$$\mathcal{L} = \sum_i (\mathcal{L}^i_{sdf} + \mathcal{L}^i_{eik} + \mathcal{L}^i_{emb} + \mathcal{L}^i_{lmk_g} + \mathcal{L}^i_{lmk_c} + \mathcal{L}^i_{res}). \tag{16}$$

在测试阶段，对于每个由j索引的3D人脸，我们最小化以下目标函数以获得其潜在嵌入和重建的3D人脸：

$$\arg\min_{\mathbf{z}_{exp}, \mathbf{z}_{id}} \sum_j (\mathcal{L}^j_{sdf} + \mathcal{L}^j_{eik} + \mathcal{L}^j_{emb}). \tag{17}$$

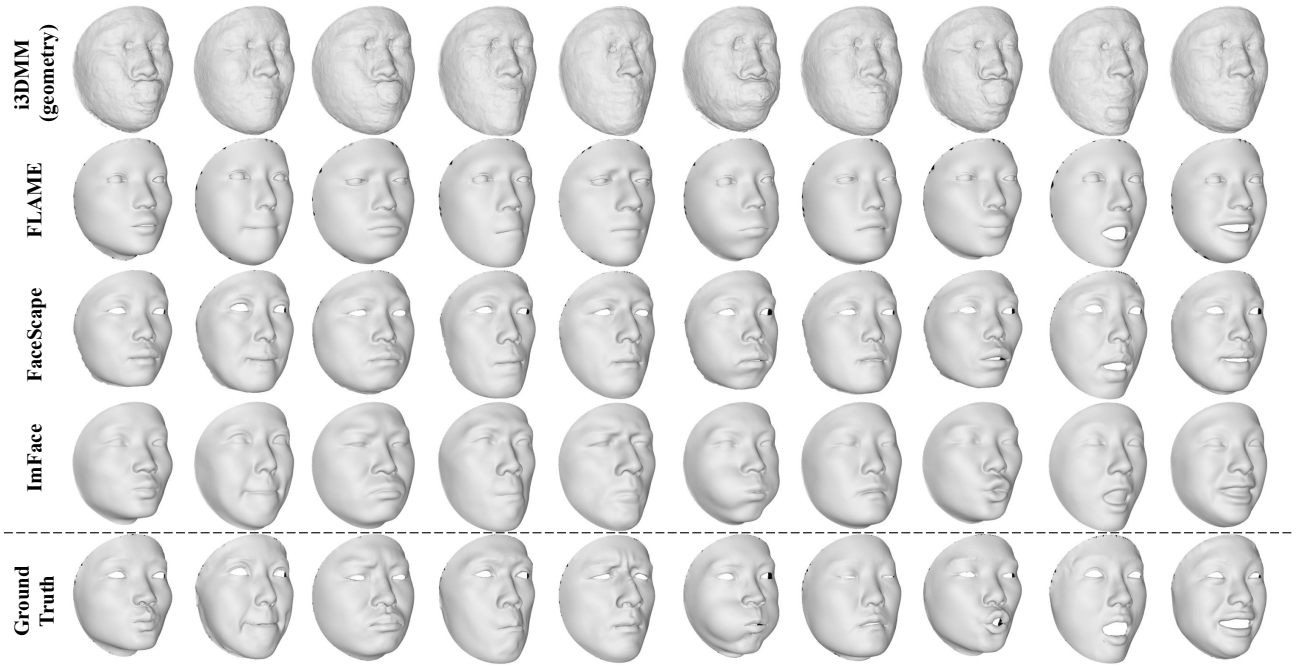


Figure 4. Reconstruction comparison with i3DMM [59], FLAME [31], and FaceScape [58]. Each column corresponds to a person with a non-neutral expression. Visually inspected, ImFace captures much richer shape variations with more compact latent embeddings.

3.5. Data Preprocessing

Since neural networks excel in fitting functions that are differentiable everywhere, the current studies on implicit functions generally require watertight input. Although functions like UDF do not demand watertightness, they are nondifferentiable when crossing a surface and are not so competent at handling details (see Fig. 3 for an illustration). We present an effective preprocessing pipeline, which generates pseudo watertight faces and defines a general SDF on them, so that the geometry and correspondence can be learned as exquisitely as on watertight objects.

Pseudo watertight face generation. The faces are rigidly aligned to frontal using landmarks and each mesh is normalized to a unit of 10 cm. The coordinate origin is set at the point 4 cm behind the nose tip, and the sphere with a radius of 10 cm is then defined as the sampling area where the mesh triangles outside are cropped away. The Ray-Triangle Intersection Algorithm [37] is applied to remove the hidden surface such as nasal and oral cavity, and the Delaunay Triangulation Algorithm [29] is then performed on x-y coordinates for an oriented and pseudo watertight mesh.

SDF computation on facial surfaces. With the pseudo watertight faces generated, SDF values can then be computed through a distance transform on them. The sign of the samples is determined simply by the angle between its distance vector to the nearest surface and the z-axis positive direction. The values of coordinates behind the facial surface are defined to be negative. We uniformly sample 250,000 points on each facial surface and 15,000 points in the sphere and calculate their signed distance and gradient vectors. The sampled data are eventually formulated as $\{(\mathbf{p}, \bar{\mathbf{n}}, \bar{s})\}$ triplets (query point, gradient vectors, signed distance value) for ImFace training.

4. Experiments

We conduct extensive subjective and objective evaluations on ImFace, and ablation studies are performed to validate the specifically designed modules.

Dataset. FaceScape [58] is a large-scale high quality 3D face Dataset consisting of 938 individuals with 20 types of expressions. The data from 365 individuals are publicly available and we mainly use them for experiments. To be specific, 5,323 face scans from 355 persons with 15 expressions are sampled as the training set, and another 200 face scans from the remaining 10 persons with 20 expressions are used as the testing set.

Network Architecture. All Mini-Nets blocks $\boldsymbol{\psi}_n$ are implemented as MLPs with 3 hidden layers and 32 dimensional hidden features activated by sine. The Hyper Nets ϕ_n are 3-layer MLPs activated by ReLU, where the hidden layer dimensionality is 64. Landmark-Nets η and η' have three 128 dimensional fully connected layers. Please refer to the supplementary material for more details of the networks.

Implementation Details. The model is trained with Adam in an end-to-end manner. We train the model for 1,500 epochs with an initial learning rate of 0.0001, and after 200 epochs, we decay it by a factor of 0.95 for every 10 epochs.

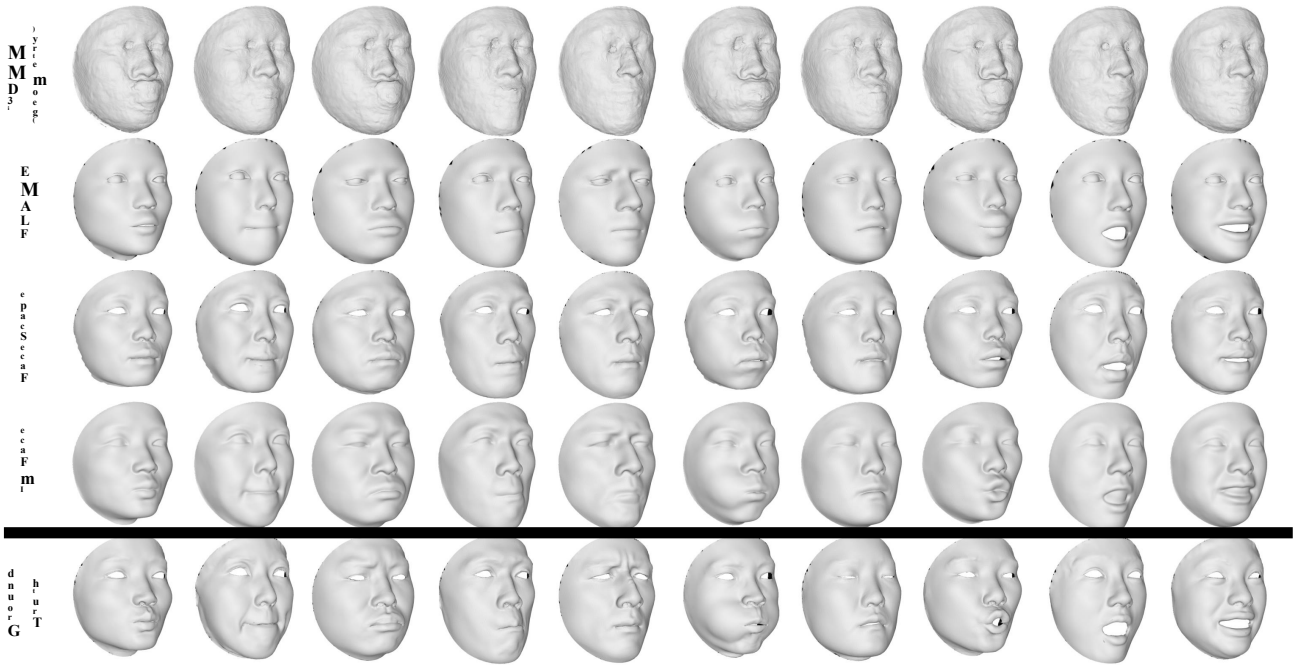


图4：与i3DMM [59]、FLAME [31]和FaceScape [58]的重建对比。每一列对应一个带有非中性表情的人物。经目视检查，ImFace以更紧凑的潜在嵌入捕获了更丰富的形状变化。

3.5. 数据预处理

由于神经网络擅长拟合处处可微的函数，当前关于隐式函数的研究通常要求输入是水密的。尽管像UDF这样的函数不要求水密性，但它们在穿越曲面时不可微，并且在处理细节方面能力不足（见图3的示意）。我们提出了一种有效的预处理流程，该流程生成伪水密面并在其上定义通用SDF，从而使得几何形状和对应关系能够像在水密物体上一样被精细地学习。

伪水密网格生成。利用关键点将人脸刚性对齐至正面，并将每个网格归一化为10 cm单位。坐标原点设定在鼻尖后方4 cm处，随后定义半径为10 cm的球体作为采样区域，裁剪掉区域外的网格三角形。采用射线-三角形相交算法[37]去除鼻腔和口腔等隐藏表面，再对x-y坐标执行Delaunay三角剖分算法[29]，生成具有方向性的伪水密网格。

面部表面上的SDF计算。在生成伪水密面片后，可通过距离变换计算SDF值。样本的符号由其到最近表面的距离向量与z轴正方向之间的夹角简单确定。面部表面后方的坐标值定义为负值。我们在每个面部表面上均匀采样250,000个点，并采样15,000个点。

在球体内采样并计算其带符号距离和梯度向量。采样数据最终被整理为$\{(\mathbf{p}, \bar{\mathbf{n}}, \bar{s})\}$三元组（查询点、梯度向量、带符号距离值），用于ImFace训练。

4. 实验

我们对ImFace进行了广泛的主观和客观评估，并进行了消融研究以验证专门设计的模块。

数据集。FaceScape [58]是一个大规模高质量的三维人脸数据集，包含938个个体的20种表情。其中365个个体的数据公开可用，我们主要使用这些数据进行实验。具体而言，从355个人的15种表情中采样了5,323个面部扫描作为训练集，另外从剩余10个人的20种表情中选取200个面部扫描作为测试集。

网络架构。所有Mini-Nets块$\boldsymbol{\psi}_n$均实现为具有3个隐藏层和32维隐藏特征的MLP，激活函数为正弦函数。Hyper Netsϕ_n为3层MLP，激活函数为ReLU，隐藏层维度为64。Landmark-Netsη和η'具有三个128维全连接层。更多网络细节请参阅补充材料。

实现细节。模型使用Adam以端到端方式训练。我们训练模型1,500个周期，初始学习率为0.0001，在200个周期后，每10个周期将学习率衰减0.95倍。

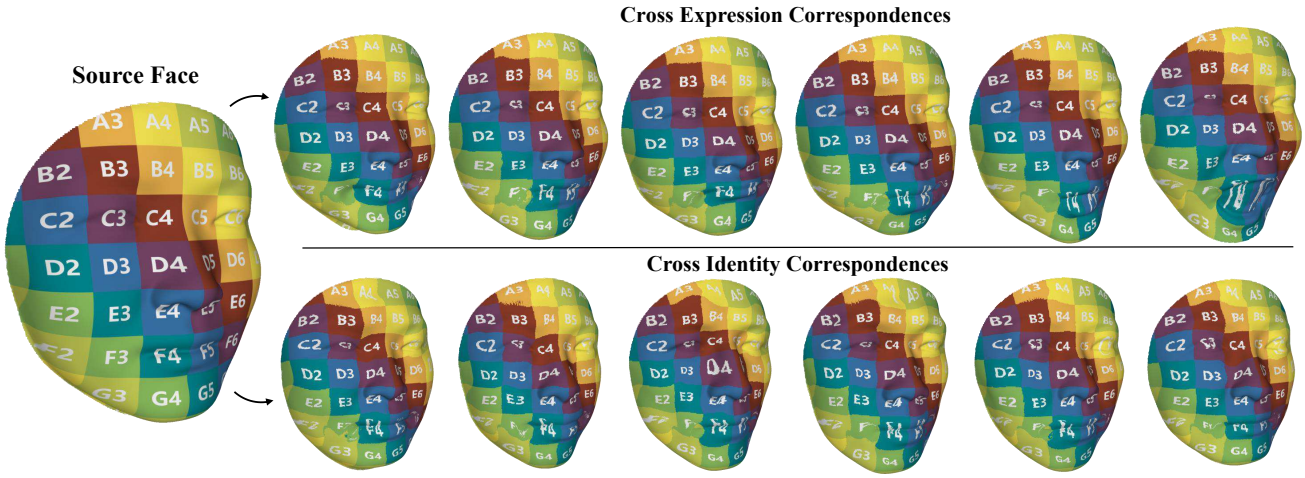


Figure 5. Correspondence results. The leftmost face is morphed into multiple expressions (upper row) and identities (bottom row).

Metrics	Dim.	Chamfer(mm)†	F-score@0.001¶
i3DMM [59]	256	1.635	42.26
FLAME [31]	400	0.971	64.73
FaceScape [58]	352	0.929	67.09
ImFace	256	**0.625**	**91.11**

Table 1. Quantitative comparison with the state-of-the-art methods (†Lower is better; ¶Higher is better).

The training phase takes around 2 days on 4 NVIDIA RTX 3090 GPUs with minibatches of size 72. During testing, it takes about 4 hours to optimize 200 samples on a single GPU.

4.1. Reconstruction

We use the proposed ImFace model to fit face scans by optimizing Eq. (17) and compare the reconstruction results with FLAME [31], FaceScape [58] and the geometry model of i3DMM [59], which signify the state-of-the-art. The official code of FLAME is used to fit the full face scans in the test set, with 300 identity parameters and 100 expression parameters. For FaceScape, we use their released bilinear model built from 938 individuals for testing, where the identity and expression parameters are 300 and 52, respectively. Note that our test scans are included in the training set of FaceScape. In addition, we modify its official code to fit the full scans instead of only the landmarks for improved results. For i3DMM, since the original model is trained only on 58 individuals, we therefore re-train the model on the same training set as in ImFace for fair comparison. In both i3DMM and ImFace, the identity and expression embeddings are 128-dimensional.

Qualitative Evaluation. Fig. 4 visualizes the reconstruction results achieved by different models, where each column corresponds to a test person with a non-neutral expression. The results also include unseen expressions during learning. i3DMM is the first deep implicit model for human heads, but it is less capable of capturing complicated deformations and fine-grained details under a relatively intricate circumstance, resulting in artifacts on the reconstructed faces. FLAME is able to well present identity characteristics, but is not so competent at dealing with nonlinear deformations, delivering stiff facial expressions. FaceScape performs more favourably mainly due to the high-quality training scans and the test faces are included in the training set, but it still cannot precisely present expression morphs. Comparatively, ImFace reconstructs faces with more accurate identity and expression properties, and it is able to preserve subtle and rich nonlinear facial muscle deformations such as frowns and pouts by fewer latent parameters.

Quantitative Evaluation. To make fair comparison, all faces are processed as described in Sec. 3.5 to remove the inner structures like eyeballs, so that the quantitative metrics can be computed in the same facial region for all the models. Specifically, the symmetric Chamfer distance and F-score are used as metrics, and the threshold of F-score is set to 0.001 as a strict standard. The results are shown in Table 1. As we can see, ImFace exceeds the counterparts by a large margin under both the metrics, which clearly validates its effectiveness.

4.2. Correspondence

In contrast to existing methods that generally require accurate face registration, correspondences can be automatically learned in INRs models. We further design training critic to enhance such a feature, and this evaluation aims to confirm it. Given two 3D faces, we use ImFace to fit them and deform the dense sampled points to the template space so that point-to-point correspondences can be fulfilled by nearest neighbor search. Fig. 5 visualizes some results generated by our method, where we manually paint color patterns on the shapes to better check the quality. In gen-

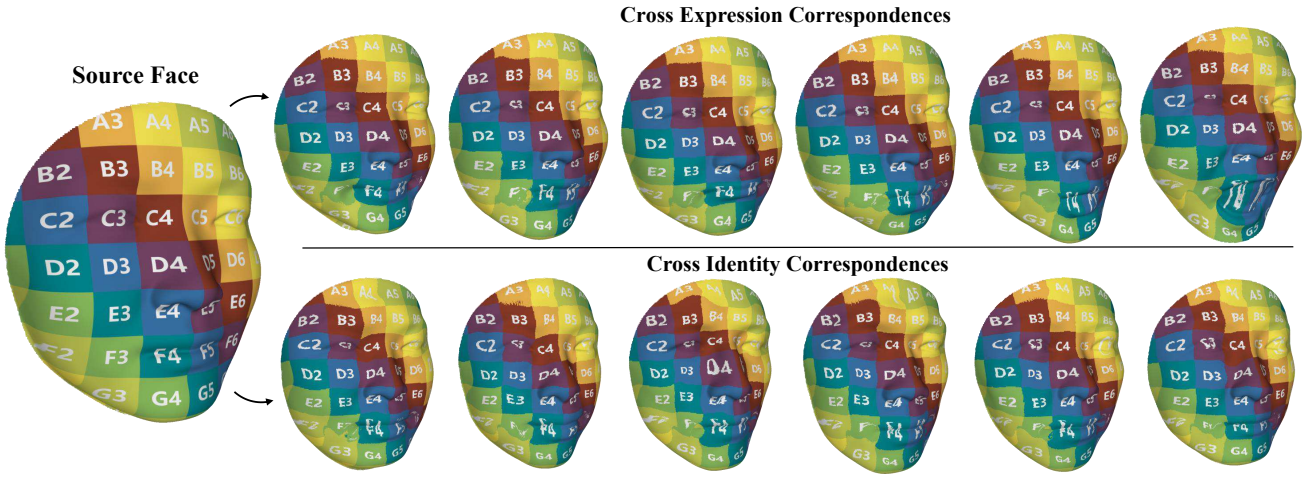


图5. 对应结果。最左侧的人脸被变形为多种表情（上行）和身份（下行）。

Metrics	Dim.	Chamfer(mm)†	F-score@0.001¶
i3DMM [59]	256	1.635	42.26
FLAME [31]	400	0.971	64.73
FaceScape [58]	352	0.929	67.09
ImFace	256	**0.625**	**91.11**

表1. 与最先进方法的定量比较（†越低越好；¶越高越好）。

训练阶段在4块NVIDIA RTX 3090 GPU上使用大小为72的小批量进行，耗时约2天。测试阶段，在单块GPU上优化200个样本约需4小时。

4.1. 重构

我们使用所提出的ImFace模型通过优化式(17)来拟合人脸扫描，并将重建结果与FLAME [31]、FaceScape [58]以及i3DMM [59]的几何模型进行比较，这些模型代表了当前最先进水平。使用FLAME的官方代码对测试集中的完整人脸扫描进行拟合，其中包含300个身份参数和100个表情参数。对于FaceScape，我们使用其发布的基于938个个体构建的双线性模型进行测试，其中身份参数和表情参数分别为300和52。需要注意的是，我们的测试扫描包含在FaceScape的训练集中。此外，我们修改了其官方代码以拟合完整扫描而非仅拟合关键点，以获得更好的结果。对于i3DMM，由于原始模型仅在58个个体上训练，因此我们在与ImFace相同的训练集上重新训练该模型，以确保公平比较。在i3DMM和ImFace中，身份和表情嵌入均为128维。

定性评估。图4可视化了不同模型所取得的重建结果，其中每一列对应一位带有非中性表情的测试者。结果还包含了训练期间未见过的表情。

学习。ImFace是首个用于人头的深度隐式模型，但在相对复杂的情况下，它捕捉复杂形变和精细细节的能力较弱，导致重建的面部出现伪影。FLAME能够很好地呈现身份特征，但在处理非线性形变方面不够擅长，导致面部表情僵硬。FaceScape表现更优主要得益于高质量的训练扫描，且测试人脸包含在训练集中，但它仍无法精确呈现表情变形。相比之下，ImFace以更准确的身份和表情属性重建面部，并且能够通过更少的潜在参数保留细微而丰富的非线性面部肌肉形变，如皱眉和噘嘴。

定量评估。为公平比较，所有面部均按第3.5节所述进行处理，去除眼球等内部结构，以便所有模型能在相同面部区域计算定量指标。具体而言，采用对称Chamfer距离和F-score作为指标，并将F-score阈值设为0.001作为严格标准。结果如表1所示。可以看出，ImFace在这两项指标上均大幅超越对比方法，这明确验证了其有效性。

4.2. 对应关系

与现有通常需要精确人脸配准的方法不同，INRs模型中可以自动学习对应关系。我们进一步设计了训练评判器以增强这一特性，本评估旨在验证这一点。给定两个3D人脸，我们使用ImFace对其进行拟合，并将密集采样点变形到模板空间，从而通过最近邻搜索实现点对点对应。图5展示了我们方法生成的部分结果，其中我们手动在形状上绘制颜色图案以更好地检查质量。总体而



(a) w/o dist. (b) w/o blend (c) w/o extend. (d) ImFace (e) GT

Figure 6. Qualitative ablation study results.

Metrics	Chamfer (mm) †	F-score@0.001 ¶
Ours w/o dist.	0.772	82.70
Ours w/o blend	0.767	82.37
Ours w/o extend.	0.705	86.98
ImFace	**0.625**	**91.11**

Table 2. Quantitative ablation study results.

eral, ImFace is able to establish pleasing correspondences across various expressions and identities. Meanwhile, it can be inspected that tiny internal texture dispersion indeed occasionally occurs around mouth corners, and this is mainly because facial shapes change drastically in these local areas under different expressions, which is also very hard case to specialized correspondence methods.

4.3. Ablation Study

ImFace is built on the following core components: disentangled deformation fields (dist.), Neural Blend-Field (blend), and improved expression embedding learning (extend.). We experimentally verify the credits of such designs.

On Disentangled Deformation Fields. To highlight the disentangled deformation learning process, we build a baseline network which contains only one deformation field to learn face shape morphs universally. Accordingly, $\mathbf{z}_{exp}$ and $\mathbf{z}_{id}$ are concatenated as the input of the hyper net. Fig. 6 (a) provides a demonstration. In spite of some fine-grained details brought by other designs, there exists a chaos on the reconstructed faces, especially for the ones with large expressions. The quantitative results in Table 2 also indicate the significance of decoupled deformation learning.

On Neural Blend-Field. We replace the Neural Blend-Field in $\mathcal{E}$, $\mathcal{I}$, $\mathcal{T}$ with vanilla MLPs of the same amount of parameters, which directly predict global deformations or SDF values of an entire face. As shown in Fig. 6 (b), a visible blur appears due to the limited capability in learning high-frequency details. The quantitative results in Table 2 confirm the necessity of Neural Blend-Field in learning sophisticated representations.

On Improved Embedding Learning. This strategy is introduced to learn more diverse and fine-grained facial deformations. As shown in Fig. 6 (c), when restricting the number of expression embeddings to be the same as expression categories, the generated expressions tend to be average. Moreover, for exaggerated expressions, such as mouth stretch, the counterpart models can hardly converge to a reasonable state.

5. Discussion

Limitations. Despite the significant advance in 3D facial shape representation, we mainly focus on face geometry modeling, whereas facial texture is less considered. Indeed, a basic texture model can be achieved by plugging a color field, but a more comprehensive INR-based 3DMM which presents the facial appearance with realistic diffuse and specular albedo remains to be explored.

Societal Impact. Our model aims at high-quality face modeling, and similar to existing models, it has the potential to be applied to downstream scenarios such as 2D reconstruction and face animation, which may result in unethical practices like privacy invasion or identity fraud. We encourage researchers and developers to consider the questions, such as how to prevent personal face data from being maliciously accessed, before applying the model to real world.

6. Conclusion

This paper presents a novel nonlinear 3D morphable face model with INRs, *i.e*. ImFace, which learns complex facial shape variations by two explicitly disentangled deformation fields associated with expression and identity, respectively, along with an improved embedding learning strategy to allow more fine-grained expressions. To precisely capture detailed facial deformations and geometries, it further presents a Neural Blend-Field. In addition, an effective preprocessing pipeline is presented, which enables INRs to work with non-watertight facial surfaces for the first time. Experiments show that ImFace is competent for this issue and outperforms the state-of-the-art counterparts.

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(a) 无分布 (b) 无混合 (c) 无扩展 (d) ImFace (e) GT

图6. 定性消融研究结果。

Metrics	Chamfer (mm) †	F-score@0.001 ¶
Ours w/o dist.	0.772	82.70
Ours w/o blend	0.767	82.37
Ours w/o extend.	0.705	86.98
ImFace	**0.625**	**91.11**

表2. 定量消融研究结果。

总体而言，ImFace能够在不同表情和身份之间建立令人满意的对应关系。同时，可以观察到，嘴角周围偶尔会出现微小的内部纹理分散，这主要是因为不同表情下这些局部区域的面部形状变化剧烈，这对于专门的对应方法来说也是极具挑战的情况。

4.3. 消融实验

ImFace基于以下核心组件构建：解耦变形场（dist.）、神经混合场（blend）以及改进的表情嵌入学习（extend.）。我们通过实验验证了这些设计的贡献。关于解耦变形场。为突出解耦变形学习过程，我们构建了一个仅包含单一变形场以通用学习面部形状变形的基线网络。相应地，$\mathbf{z}_{exp}$和$\mathbf{z}_{id}$被拼接作为超网络的输入。图6（a）给出了演示。尽管其他设计带来了一些细粒度细节，但重建面部仍存在混乱，尤其是对于具有大表情的面部。表2中的定量结果也表明了解耦变形学习的重要性。

关于神经混合场。我们将$\mathcal{E}$、$\mathcal{I}$、$\mathcal{T}$中的神经混合场替换为参数数量相同的普通MLP，直接预测整个面的全局形变或SDF值。如图6(b)所示，由于学习能力有限，出现了明显的模糊。高频细节。表2中的定量结果证实了神经混合场在学习复杂表示中的必要性。

关于改进嵌入学习。引入该策略是为了学习更多样化、更细粒度的面部形变。如图6(c)所示，当将表情嵌入的数量限制为与表情类别数量相同时，生成的表情往往趋于平均化。此外，对于夸张的表情，如嘴巴拉伸，对应的模型难以收敛到合理状态。

5. 讨论

局限性。尽管三维面部形状表示取得了显著进展，我们主要关注面部几何建模，而对面部纹理的考虑较少。实际上，通过引入颜色场可以实现基本的纹理模型，但一个更全面的基于INR的三维形变模型，能够以逼真的漫反射和高光反射率呈现面部外观，仍有待探索。

社会影响。我们的模型旨在实现高质量的人脸建模，与现有模型类似，它有可能应用于下游场景，如二维重建和人脸动画，这可能导致隐私侵犯或身份欺诈等不道德行为。我们鼓励研究人员和开发者在将模型应用于现实世界之前，考虑如何防止个人人脸数据被恶意获取等问题。

6. 结论

本文提出了一种新颖的基于INR的非线性3D可变形人脸模型，即ImFace，该模型通过两个分别与表情和身份相关联的显式解耦变形场，以及一种改进的嵌入学习策略，来学习复杂的面部形状变化，从而支持更细粒度的表情。为了精确捕捉细致的面部变形和几何结构，本文进一步提出了一种神经混合场。此外，还提出了一种有效的预处理流程，首次使INR能够处理非水密的人脸表面。实验表明，ImFace能够胜任这一问题，并且优于现有的最先进方法。

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